COVID-19 GLOBAL IMPACT DASHBOARD: EXECUTIVE SUMMARY

Author: Sabiya Ansari | Role: Data Analytics Intern

Dataset Source: Our World in Data (OWID)

1. PROJECT OBJECTIVE

To analyse global COVID-19 time-series data (cases, deaths, and population metrics), clean real-world anomalies, derive key epidemiological indicators (7-day rolling averages, Case Fatality Rates), and communicate findings visually through an interactive Power BI dashboard.

2. METHODOLOGY & DATA CLEANING

- Cleaned 44,000+ rows of messy time-series data using Python (Pandas).
- Handled missing daily records by substituting zero values where appropriate.
- Filtered out global/regional aggregate groups to ensure accurate country-to-country comparisons.
- Plotted new metrics including 7-day case rolling averages to eliminate reporting noise.

3. KEY FINDINGS

- Raw case counts favour populous countries falsely. Population-normalized metrics (cases per million) revealed that nations like the US, Peru, and Brazil experienced the highest relative impact density.
- Raw daily counts exhibited artificial weekend dips; 7-day smoothing successfully exposed true epidemiological wave patterns.
- Case Fatality Rates varied substantially across regions due to testing capacity differences, demographic makeup, and healthcare infrastructure readiness.

4. CONCLUSION & RECOMMENDATIONS

The interactive Power BI dashboard successfully empowers non-technical stakeholders to compare regional pandemic trajectories instantly. Future analyses should incorporate vaccination rollout speed to evaluate immunization effectiveness against severe outcomes.